Comparative Analysis of Nature-Inspired Designs in Plate Heat Exchangers

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Abstract- ***In this study, a biomimetic approach is adopted to further enhance the thermal performance of PHEs. Inspired by natural patterns, four different plate designs—flat plate, ribbed plate, vein-patterned plate, and a leaf-structured plate—were developed and analyzed using computational fluid dynamics (CFD). The baseline flat plate design was validated against existing data to ensure accuracy, following which detailed simulations were conducted for all configurations. Key performance parameters such as convective heat transfer coefficient, pressure drop, and overall heat transfer rate (Q) were calculated for each design. Among the four, the leaf-structured plate demonstrated superior performance, achieving higher heat transfer rates with relatively lower pressure drop. This improvement is attributed to the efficient fluid distribution and turbulence generation provided by the leaf-inspired geometry. The results highlight the potential of nature-inspired designs in advancing the efficiency of compact heat exchangers.***

Keywords - Flat Plate Heat Exchanger, Biomimicry, CFD, Heat Transfer, Pressure Drop, Nature-Inspired Design

I. Introduction

Flat plate heat exchangers (PHEs) are widely used in various thermal systems due to their compact structure, high heat transfer efficiency, and ease of maintenance. While studying or designing PHEs, the primary objective is always to maximize the heat transfer coefficient while minimizing the pressure drop to ensure optimal performance.

Biomimicry has emerged as a powerful paradigm in thermal management, leveraging nature-inspired designs to optimize heat transfer, reduce drag, and enhance energy efficiency across various engineering applications. By emulating natural structures and mechanisms that have evolved for thermal regulation and fluid flow optimization, researchers have developed innovative heat exchange systems and materials that outperform conventional designs.

One significant area of advancement lies in the enhancement of convective and evaporative heat transfer through surface texturing inspired by biological forms. Wang et al. (2021) demonstrated that structures modeled after shark skin, lotus leaves, and butterfly scales can improve heat transfer rates by up to 25% while

simultaneously reducing drag by 10–20% under turbulent flow conditions [1]. These micro- and nanoscale textures manipulate boundary layer behaviour, creating more favourable flow conditions for thermal exchange.

Similarly, Rupp and Gruber (2019) highlighted how the intricate geometry of plant leaves, especially sun leaves with dissected margins, contributes to enhanced convective and evaporative cooling [2]. These findings align with the work of Zuazua-Ros et al. (2019), who explored the integration of bio-inspired surface morphologies into architectural façades, demonstrating improved heat rejection capabilities in building cooling cycles [3]. Such passive strategies represent promising avenues for reducing energy consumption in the built environment.

The role of structural morphology is further underscored in studies examining marine and avian biomimicry. Smith et al. (2019) revealed that heat exchanger designs inspired by oyster shells, particularly those with toothed edge features, can facilitate localized cooling and achieve temperature reductions of up to 10°C below ambient levels [4]. Gadakh (2022) adds to this by comparing a bird feather-inspired plate heat exchanger with traditional chevron-type designs. While the biomimetic model has a slightly lower heat transfer rate, it offers an 8% reduction in pressure drop, thereby enhancing overall flow efficiency—a critical consideration in system optimization [7].

The manipulation of fluid flow within confined channels also benefits from nature-inspired geometry. Ebrahimi and Naranjani (2018) reported a remarkable 277.9% enhancement in heat transfer when introducing pyramidal protrusions in flat-plate channels, illustrating how disruption of laminar flow can significantly boost thermal performance [5]. In parallel, Jaffer and Jaffer (2024) provide predictive models for mixed convection heat transfer on roughened surfaces, confirming through experimental validation the utility of textural modifications in practical applications [6].

Beyond surface texturing and geometric design, biomimicry has also informed material innovations and phase change systems. Hu et al. (2023) introduce a tree-inspired thermal management composite that mimics vascular transport through layered EVA films, achieving directional heat distribution akin to xylem and phloem in plant systems [8]. Similarly, Anand and Sharma (2023) develop a micropillar wick structure based on the peristome of *Nepenthes alata*, enhancing thin-film evaporation through capillary action and surface energy manipulation [12].

Structural inspiration from human physiology also contributes to this field. Sousa et al. (2023) proposed an alveolus-like fin design for extended surfaces, which demonstrates a 12% improvement in heat transfer under mixed convection conditions compared to conventional counterparts [9]. In extreme environments, such as space, Telford (2023) shows that a biomimetic heat pipe, modelled to optimize capillary action and thermal conductivity, achieves a 30% enhancement over traditional designs, proving vital for thermal regulation in aerospace applications [10].

Completing the spectrum of innovations, Liu et al. (2014) presented a transformation thermodynamics-based plate heater, employing advanced functional materials to guide heat flux and maintain a uniform surface temperature—echoing the precision of heat regulation seen in biological systems [11].

Collectively, these studies reveal a compelling narrative: by closely examining and mimicking the strategies found in nature—from plant leaves and tree vasculature to marine shells and bird feathers—engineers and scientists are revolutionizing the design of heat transfer systems. This biomimetic approach not only enables performance gains in thermal regulation and fluid dynamics but also contributes to the

broader goals of energy efficiency and sustainability across a range of industries.

II. Working and Construction

A plate heat exchanger works by transferring heat between two fluid streams flowing through alternating channels formed by stacked plates. These plates are typically made of thermally conductive metals and can be designed with various surface features to improve thermal performance. The heat transfer process is maximized by increasing surface area and promoting turbulent flow, both of which contribute to higher convective heat transfer coefficients. The fluids typically flow in opposite directions (counterflow configuration), which enhances the temperature gradient and overall efficiency. The performance of a PHE is often evaluated using parameters like pressure drop, heat transfer coefficient, and compactness—defined as the heat transfer area per unit volume of the exchanger (m^2/m^3).

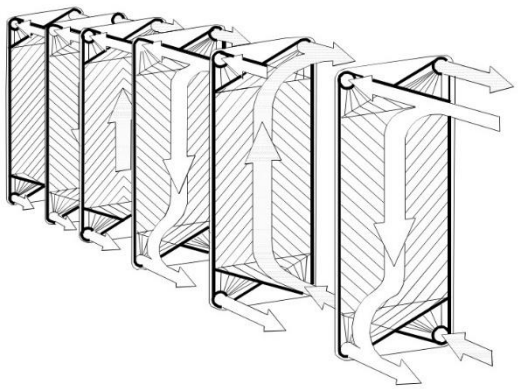


Figure 1 Arrangemnet of PHE

In this study, four distinct plate surface designs were developed and analysed using SolidWorks Simulation: a plane plate, a grooved plate, a vein-structured plate inspired by fish gills, and a leaf-structured plate inspired by the tracheal system. The geometric dimensions of each design will be provided in a later section.



Figure 2. Flat Plate HE

The plain plate serves as the simplest configuration, featuring a flat surface without any structural modifications (Figure 1). While it allows fluid to flow smoothly with minimal resistance, it lacks the surface complexity needed to generate turbulence or improve thermal mixing. As a result, this design demonstrates the lowest heat transfer efficiency among the four types, though it benefits from low pressure drop and ease of manufacturing.



Figure 3. Grooved Plate HE

To improve upon this baseline, the grooved plate design introduces shallow surface channels that disturb the boundary layer and promote mild turbulence in the flow (Figure 2). These grooves increase the heat transfer surface area and improve mixing, leading to higher convective heat transfer. The design strikes a balance between improved thermal performance and moderate pressure drop, making it a practical enhancement over the plane plate.



Figure 4 Vein type Plate HE

The vein-structured plate, inspired by the chevron-like arrangement of fish gills, incorporates 21 angled wings oriented at 30° on the plate surface (Figure 3). Each wing is 0.35 mm wide and 1.7 mm high. The first wing features a gill spacing of 1.17 mm, while the rest are spaced at 1.43 mm to avoid clogging. In areas without wings, flow routers measuring 0.48 mm in width and spaced 1.5 mm apart are incorporated to ensure smooth and uniform fluid distribution across the plate. This design mimics the natural flow guidance of fish gills, where water flows over closely spaced ridges for efficient oxygen exchange. Similarly, in the PHE, this structure generates strong secondary flows and vortices that significantly enhance turbulence and surface interaction, improving heat transfer performance. The compactness of this configuration is calculated to be 6280 m^2/m^3, indicating a well-utilized heat transfer surface relative to volume.

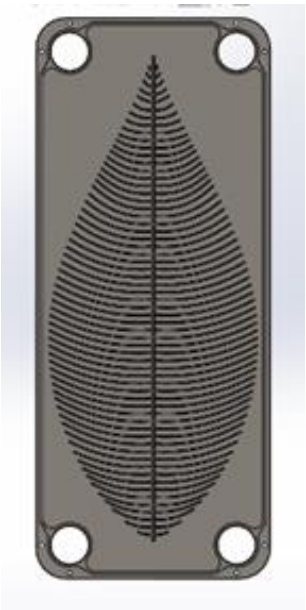


Figure 5. Leaf type Plate HE

The most complex design is the leaf-structured plate, which draws from the 3D geometry of tracheal and leaf vein systems (Figure 4). Hemispherical fins are uniformly distributed on both sides of the plate, each with a diameter of 3.60 mm and a height of 1.60 mm. These fins interrupt the fluid flow, causing chaotic mixing and improved convective heat transfer. Additionally, 16 capsule-shaped flow routers are positioned at the hot fluid inlet and outlet regions to serve as diverters. These help to evenly distribute the fluid and minimize stagnation zones or bypass flows. This biomimetic design achieves the highest compactness value of 8150 m^2/m^3 among all tested configurations, showing its superiority in packing more thermal surface area into a smaller volume.

All four designs were developed and simulated using SolidWorks, which enabled precise 3D modelling and computational analysis of flow and thermal behaviour. High-resolution meshing was applied to ensure accurate simulation of velocity fields, pressure drops, and temperature gradients. The SolidWorks Simulation environment provided a seamless transition from CAD design to performance analysis, allowing

for direct evaluation of how each geometric feature influenced the heat transfer characteristics.

Through the integration of biomimetic surface structures, significant improvements were achieved over conventional flat plate configurations. Designs inspired by natural systems such as fish gills and tracheal networks not only enhanced thermal performance but also maintained manageable pressure losses. These findings highlight the potential of nature-based engineering to create more efficient, compact, and sustainable heat exchanger systems.

Geometric parameters of Plate Heat exchangers	
Length of Heat exchanger	739 mm
Width of Heat Exchanger	309 mm
Hot fluid inlet-outlet diameter	50 mm
Cold fluid inlet-outlet diameter	50 mm
Number of exchanger plates	5

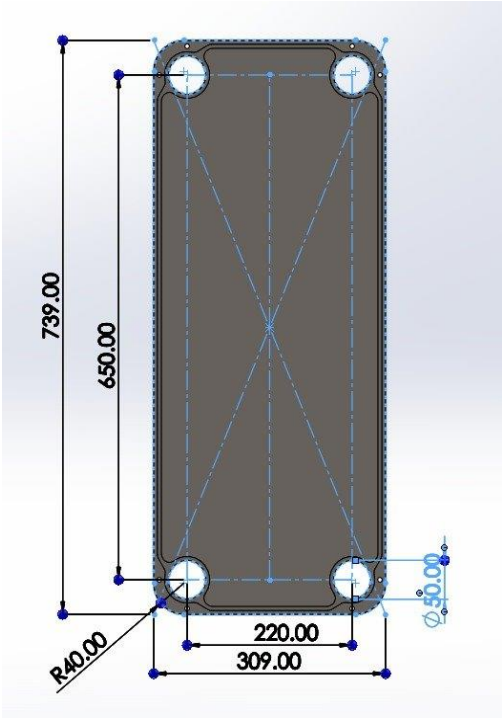


Figure 6. Geometric Dimensions

III. Materials and Methods

A. Numerical modelling

The numerical analysis for evaluating the thermal and fluid flow performance of various flat plate heat exchanger (PHE) designs was conducted using SolidWorks Flow Simulation. This CFD tool enables the simulation of internal fluid flow and heat transfer in complex geometries under steady-state conditions. Four PHE configurations—flat plate, ribbed plate, vein-patterned plate, and a leaf-structured plate—were examined to determine the impact of biomimetic geometry on heat exchanger efficiency.

The simulations were based on solving the fundamental governing equations of fluid dynamics: the continuity equation, the momentum equation, and the energy equation.

The continuity equation ensures mass conservation and is given by:

$$\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0. \tag{1}$$

Where v is the velocity vector of the fluid.

The momentum equation (Navier-Stokes) governs the conservation of momentum and is written as:

$$u\frac{\partial(\rho u)}{\partial x}+v\frac{\partial(\rho v)}{\partial y}+w\frac{\partial(\rho w)}{\partial z}=-\frac{\partial P}{\partial x} +\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right), \tag{2}$$

$$u\frac{\partial(\rho u)}{\partial x}+v\frac{\partial(\rho v)}{\partial y}+w\frac{\partial(\rho w)}{\partial z}=-\frac{\partial P}{\partial y} +\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right), \tag{3}$$

$$u\frac{\partial(\rho u)}{\partial x}+v\frac{\partial(\rho v)}{\partial y}+w\frac{\partial(\rho w)}{\partial z}=-\frac{\partial P}{\partial z} +\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right). \tag{4}$$

The energy equation accounts for thermal energy conservation and is given by:

$$u\frac{\partial(\rho T)}{\partial x}+v\frac{\partial(\rho T)}{\partial y}+w\frac{\partial(\rho T)}{\partial z}=-\frac{\partial P}{\partial x}+\frac{\mu}{T}\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right). \quad (5)$$

where C_p is the specific heat capacity at constant pressure (J/kg·K), T is the temperature (K), k is the thermal conductivity (W/m·K), and Q is the internal heat generation per unit volume (W/m³).

In SolidWorks Flow Simulation, appropriate boundary conditions were applied, including specified inlet velocity and temperature, atmospheric pressure at the outlet, and no-slip adiabatic wall conditions where required. The material properties of water were used as the working fluid. A fine mesh with local refinements near solid-fluid interfaces was generated to accurately capture flow separation and thermal gradients. The solver automatically managed the selection of laminar or turbulent flow regimes based on Reynolds number, and turbulence was handled using a modified k-ε model embedded within the software. Convergence criteria were set based on stabilization of heat transfer rate and pressure values, ensuring reliable comparison between the different plate configurations.

B. Model Validation

To ensure the accuracy and reliability of the CFD model used for simulating different plate heat exchanger (PHE) geometries, an initial validation was conducted using a plain plate heat exchanger. The objective was to confirm that the CFD model adheres to basic thermodynamic principles—particularly the conservation of energy—before applying it to enhanced biomimetic designs.

In this validation step, hot oil and cold water were used as the two working fluids. The simulation was performed under steady-state conditions, and the outlet temperatures of both fluids were extracted. These values were used to calculate the heat transferred theoretically and to compare the heat lost by the hot fluid with the heat gained by the cold fluid.

Boundary Conditions in Analysis	
Inlet Temperature of Oil	353K
Inlet Temperature of Water	308 K
Mass Flow rate of oil	0.04 kg/s
Mass flow rate of	0.02 kg/s

Input parameters	
Mass flow rate of hot fluid (oil), $\dot{m}_{hot}$	0.02 kg/s
Mass flow rate of cold fluid (water), $\dot{m}_{cold}$	0.04 kg/s
Inlet temperature of oil, $T_{in\ hot}$	79.85 °C
Outlet temperature of oil $T_{out\ hot}$	66.45 °C
Inlet temperature of water $T_{in\ cold}$	34.85 °C
Outlet temperature of water $T_{out\ cold}$	37.89 °C
Specific heat capacity of oil $C_{p\ hot}$	2131 J/kgK
Specific heat capacity of water $C_{p\ cold}$	4179.5 J/kgK

Theoretical Heat Transfer Calculations:

Heat lost by hot fluid (oil):

$$Q_{hot} = \dot{m}_{hot} * C_{p\ hot} * (T_{in\ hot} - T_{out\ hot}) \quad (6)$$

Heat gained by cold fluid (water):

$$Q_{cold} = \dot{m}_{cold} * C_{p\ cold} * (T_{out,\ cold} - T_{in,\ cold}) \quad (7)$$

Result Interpretation:

- Heat lost by hot fluid (oil): 571.11 W
- Heat gained by cold fluid (water): 508.23 W

The difference between the two values is within an acceptable margin and may arise due to simulation assumptions, fluid property approximations, and possible numerical losses. These results confirm that the CFD model successfully conserves energy and accurately predicts heat transfer performance for a plain plate configuration.

With the model validated, the same simulation framework was extended to analyze enhanced PHE surfaces such as grooved, vein-structured, and leaf-structured plates, ensuring consistency and confidence in the subsequent performance evaluation.

C. Analysis

Following the validation of the CFD model against standard flat plate heat exchanger data to ensure accuracy and reliability, further analysis was conducted on biomimetic geometries. Four plate designs—flat (validated baseline), grooved, vein-inspired, and leaf-shaped—were developed using SolidWorks with consistent outer dimensions to maintain uniformity in comparison. The geometries were imported into a CFD solver where steady-state simulations were carried out using structured meshing and appropriate boundary conditions: water flow rate at 0.04 kg/s, oil flow rate at 0.02 kg/s, water inlet temperature at 308 K, and oil inlet temperature at 353 K. Post-validation, the focus of analysis was on three critical parameters: pressure drop, heat transfer coefficient, and outlet temperature of oil. These metrics enabled a detailed evaluation of how bio-inspired modifications influence the thermal and hydraulic performance of the heat exchanger, demonstrating the potential advantages of incorporating natural design principles.

IV. Results

The CFD simulations for different plate heat exchanger (PHE) configurations—Plain Plate, Grooved (Ribbed) Plate, Vein-Structured Plate, and Leaf-Structured Plate—were analyzed in detail. The effect of geometry on fluid flow and heat transfer performance was evaluated using key visualization outputs: velocity contours, temperature contours, pressure contours, surface temperature plots, and flow trajectories. Quantitative values of heat transfer rates and pressure drops for oil and water were also computed using simulated outlet and inlet conditions.

Plain Plate Heat Exchanger

This configuration represents the simplest PHE design, with smooth, flat plates and no surface enhancement.

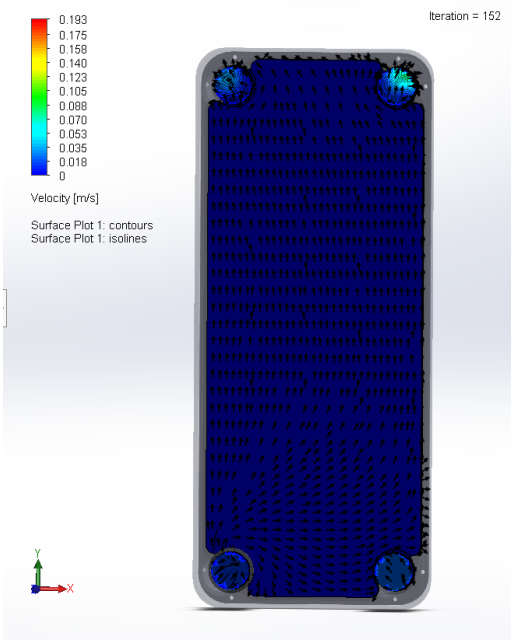


Figure 7. Velocity Contours of plain PHE

The velocity contours show a consistent, laminar flow with parallel streamlines from inlet to outlet. There is minimal disruption or swirling, indicating that flow follows a straight path with little interaction between fluid layers. This limits turbulence and, as a result, reduces the convective heat transfer coefficient.

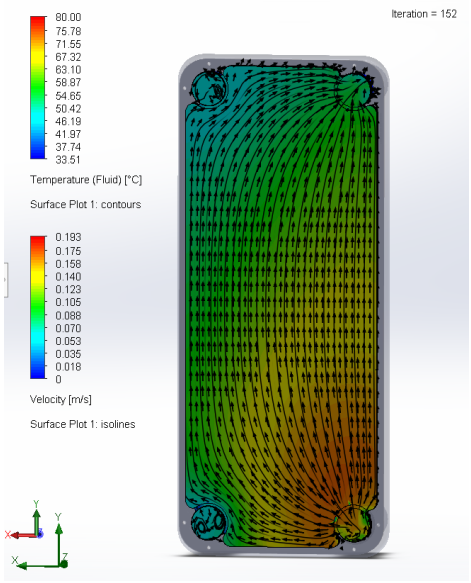


Figure 8. Temperature Contours of Plain PHE

The temperature gradient is gradual and linear from the hot to cold side. The lack of mixing means that the thermal boundary layer grows steadily, reducing heat transfer efficiency along the length of the plate.

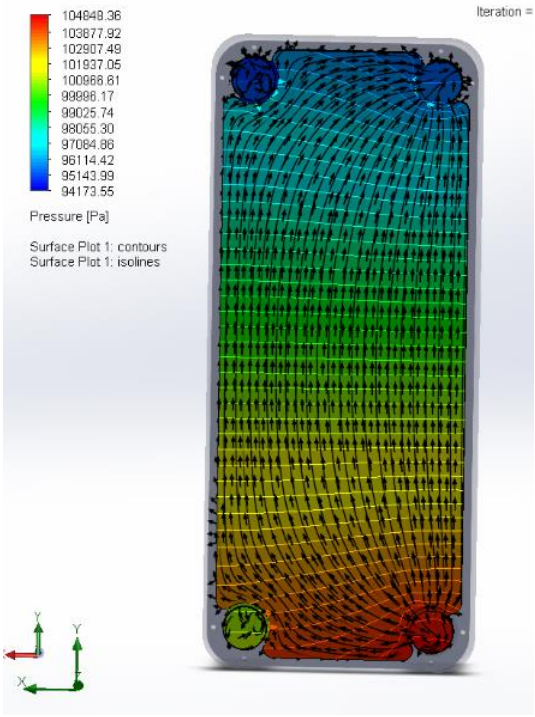


Figure 9. Pressure Contours of Plain PHE

The pressure field displays a smooth and steady drop across the flow path. The absence of geometric features results in a low-pressure loss, which is favourable for pumping efficiency but also reflects minimal flow interaction with the wall.

Surface Temperature Plot

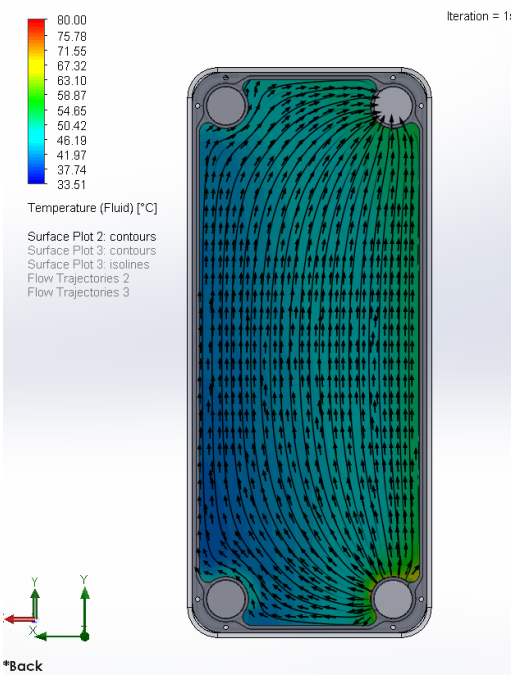


Figure 10. Surface Temperature plot of Plain PHE

The surface plot shows a moderate change in plate temperature, concentrated mainly in central regions. Outer edges remain underutilized, suggesting limited fluid contact and suboptimal thermal surface usage.

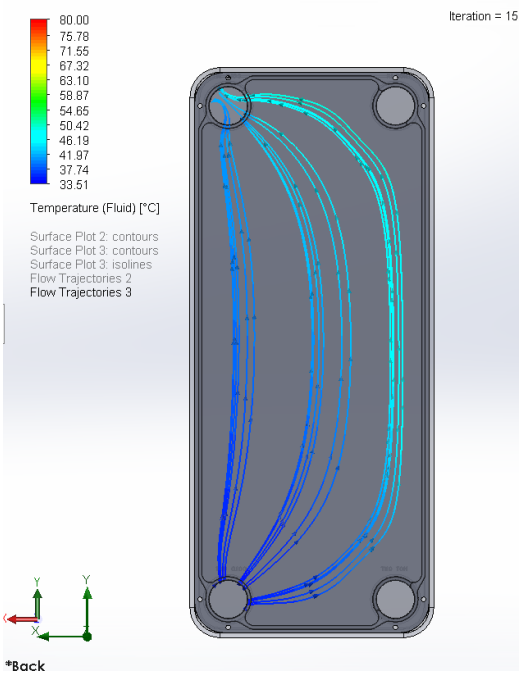


Figure 11. Flow Trajectories of Cold Fluid in Plain PHE

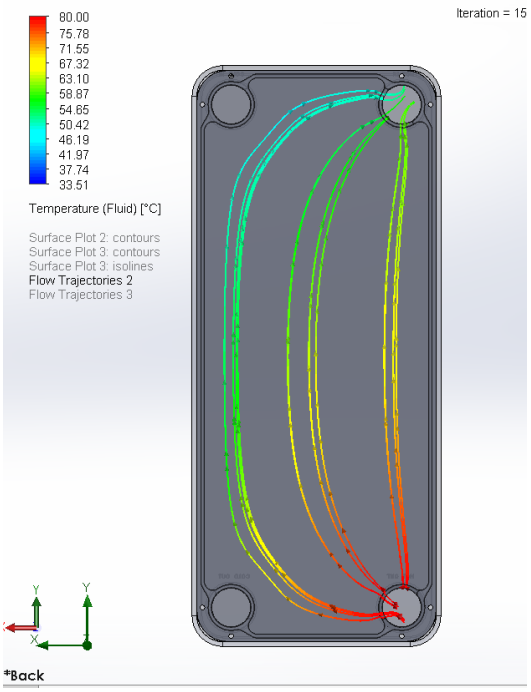


Figure 12. Flow Trajectories of Hot Fluid in Plain PHE

Flow trajectories follow straight lines from inlet to outlet, confirming laminar, unidirectional flow. No vortices or recirculation zones are observed, which means limited fluid mixing and interaction.

The plain plate shows the lowest pressure drop but also moderate heat transfer performance due to minimal turbulence.

Grooved (Ribbed) Plate Heat Exchanger-

This design incorporates transverse ribs to enhance wall turbulence and mixing.

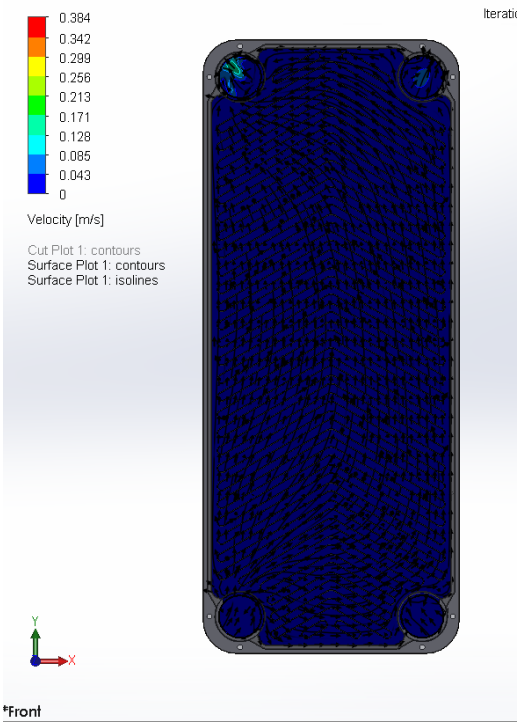


Figure 13. Velocity Contours of Grooved PHE

Velocity increases near the rib edges, creating localized zones of high-speed flow. These features promote disruption of the boundary layer and increase turbulence intensity, contributing to higher convective heat transfer.

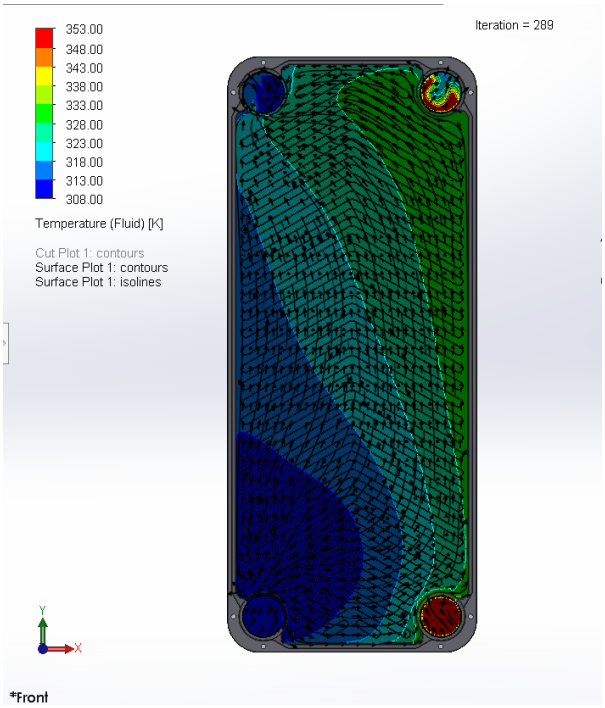


Figure 14. Temperature Contours of Grooved PHE

More uniform temperature distribution is observed along the flow direction. The temperature gradient is steeper near the ribs, indicating better local heat transfer due to increased turbulence.

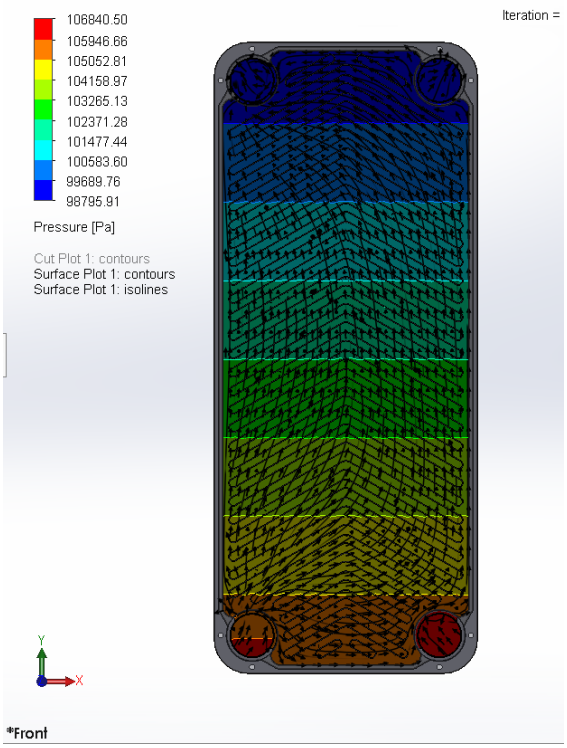


Figure 15. Pressure Contours of Grooved PHE

Moderate pressure drop is visible as fluid navigates around ribbed features. This resistance increases with rib depth and spacing but is acceptable given the improvement in thermal performance.

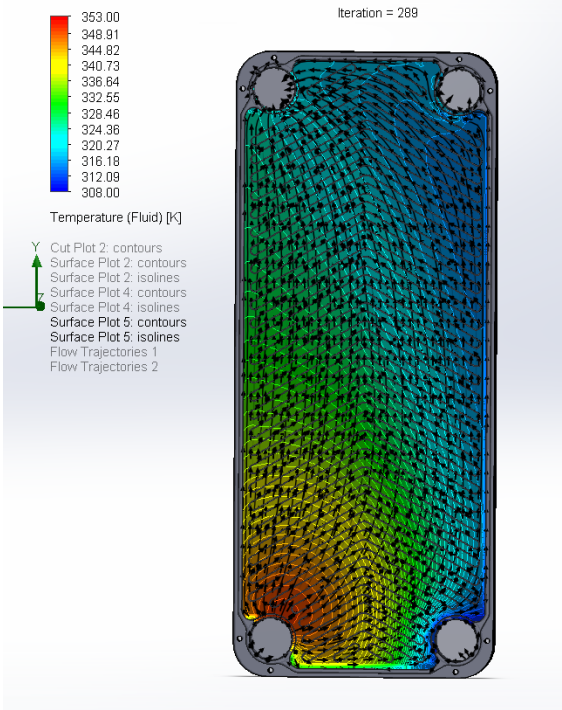


Figure 16. Surface Temperature Plot of Grooved PHE

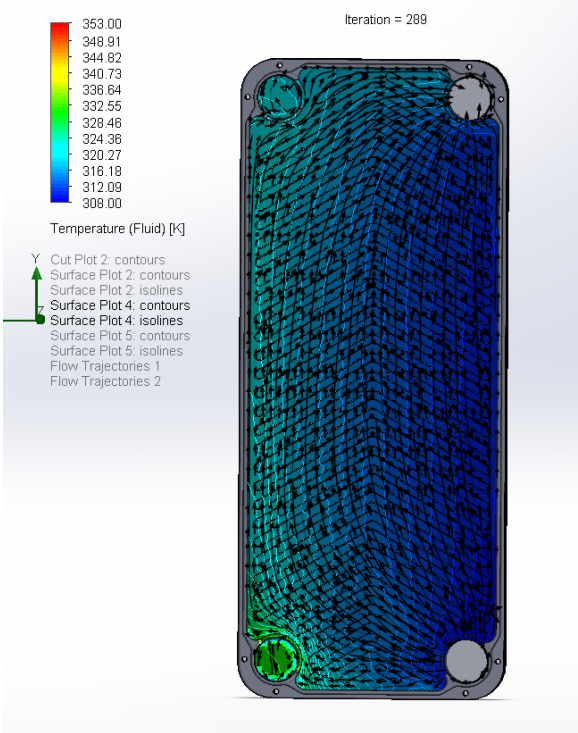


Figure 17. Surface Temperature Plot of Grooved PHE

Wall temperature is more uniformly distributed compared to the plain plate. The ribs increase the thermal contact area, enabling more effective utilization of the plate surface.

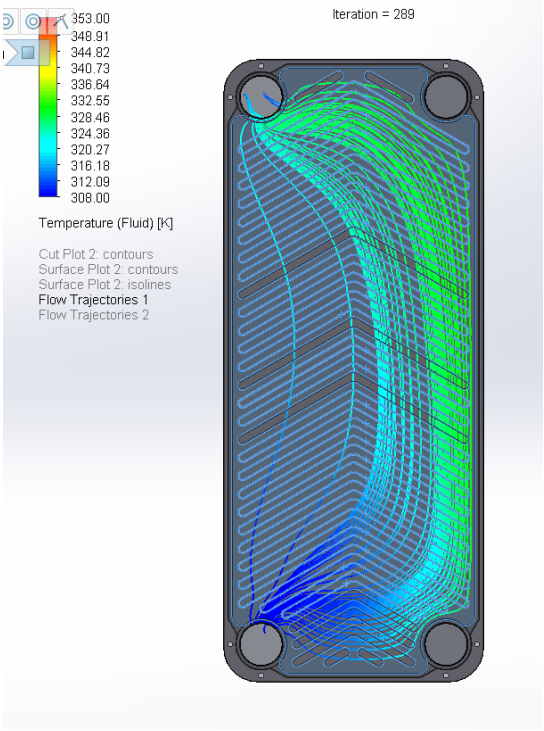


Figure 18. Flow Trajectories of cold fluid in Grooved PHE

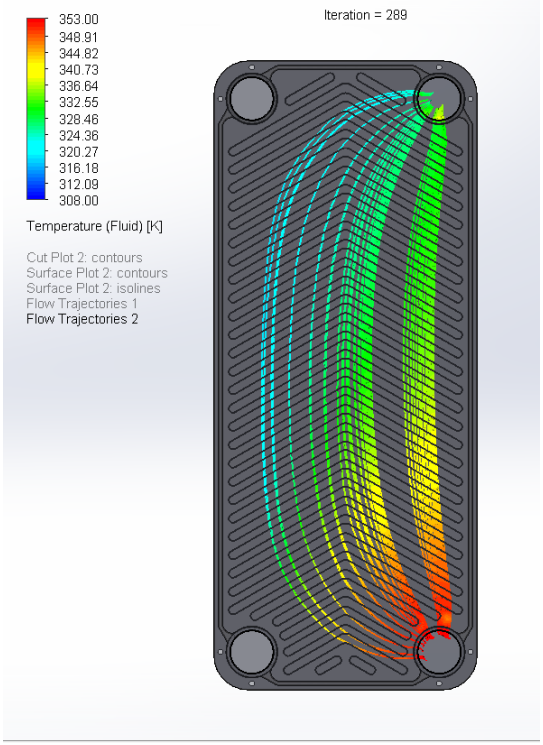


Figure 19. Flow Trajectories of hot fluid in Grooved PHE

Flow paths show deviation near the ribs, with some minor swirling and reattachment zones. This enhances mixing and disrupts the thermal boundary layer.

This design provides a good balance between enhanced heat transfer and moderate pressure losses.

Vein-Structured Plate Heat Exchanger -

Inspired by natural vein patterns, this configuration introduces curved flow paths to increase complexity and turbulence.

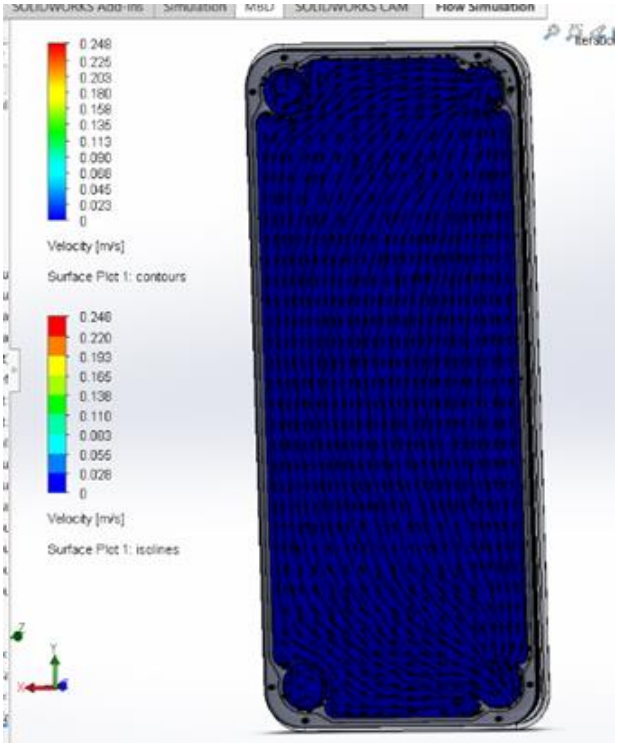


Figure 20. Velocity Contours of Vein Structured PHE

Significant flow redirection and vortex formation occur around the veins. These recirculations increase flow disturbance and improve heat transport by enhancing mixing near the wall.

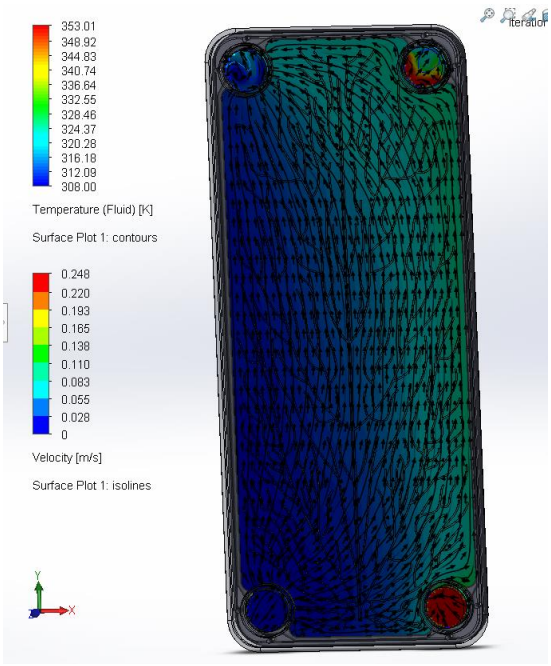


Figure 21. Temperature Contours of Vein structured PHE

Sharp temperature gradients near the veins indicate active convective heat transfer. Flow is distributed to a larger surface area due to channel guidance.

Pressure Contours

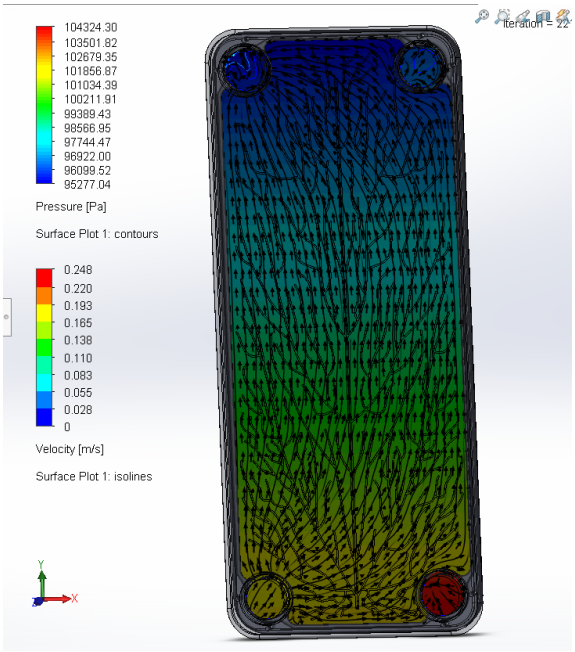


Figure 22.Pressure Contours of Vein structured PHE

The complex geometry causes a higher pressure drop, particularly where veins intersect. Flow deceleration and redirection at multiple locations contribute to the increased resistance.

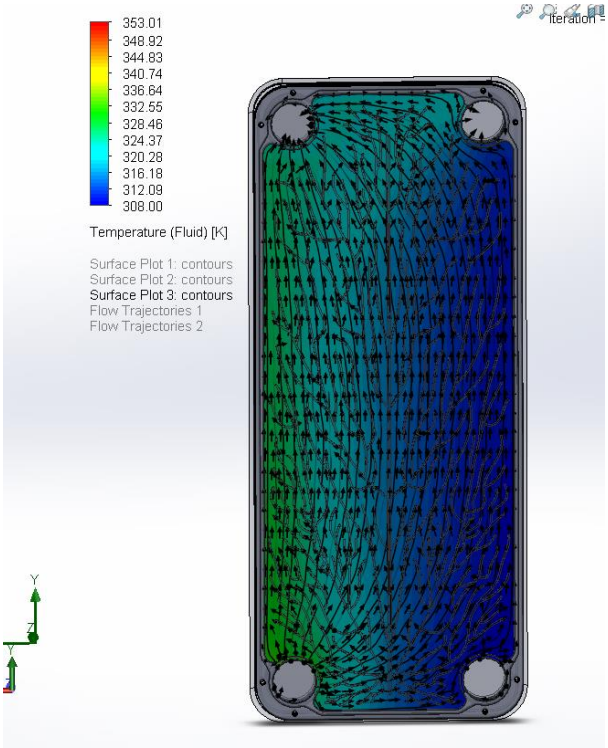


Figure 23. Surface Temperature Plot of Vein structured PHE

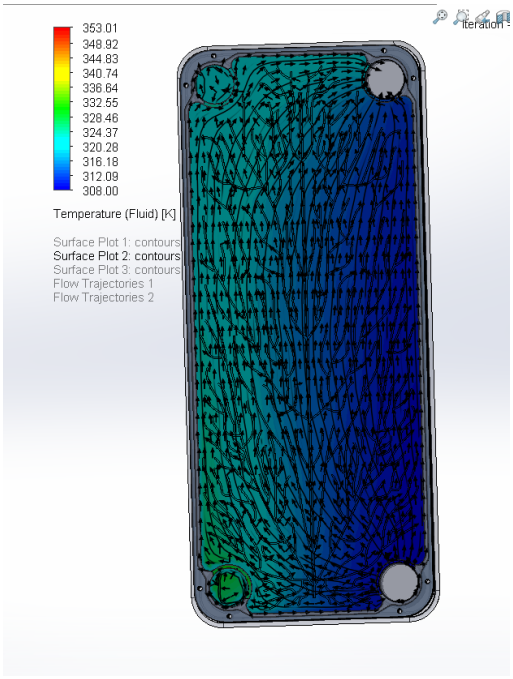


Figure 24. Surface Temperature Plot of Vein structured PHE

The plate surface shows extensive thermal activation, especially in areas near the flow-diverting veins, reflecting effective heat utilization.

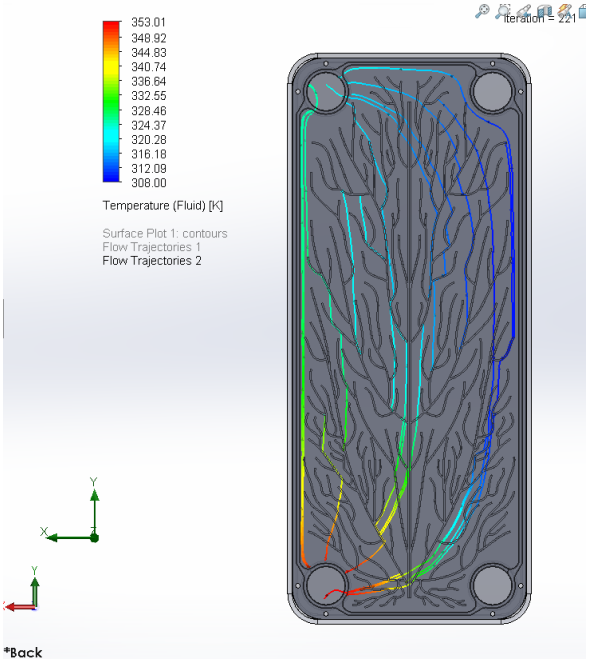


Figure 25. Flow Trajectories of hot fluid in Vein structured PHE

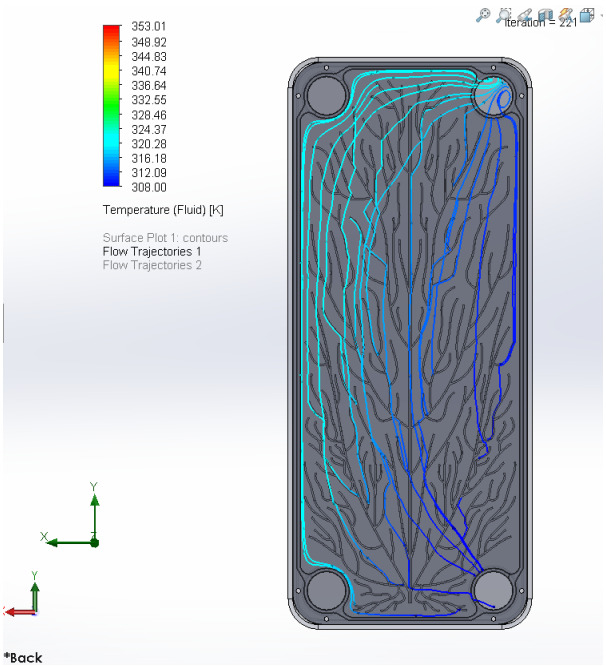


Figure 26. Flow Trajectories of cold fluid in vein structured PHE

Trajectory plots show splitting and recombination of flow streams, which significantly enhances mixing and thermal energy exchange.

This design shows strong thermal performance but at the cost of a high pressure drop on the hot fluid side.

Leaf-Structured Plate Heat Exchanger

Mimicking the distribution of veins in leaves, this model uses branching flow paths to maximize surface coverage.

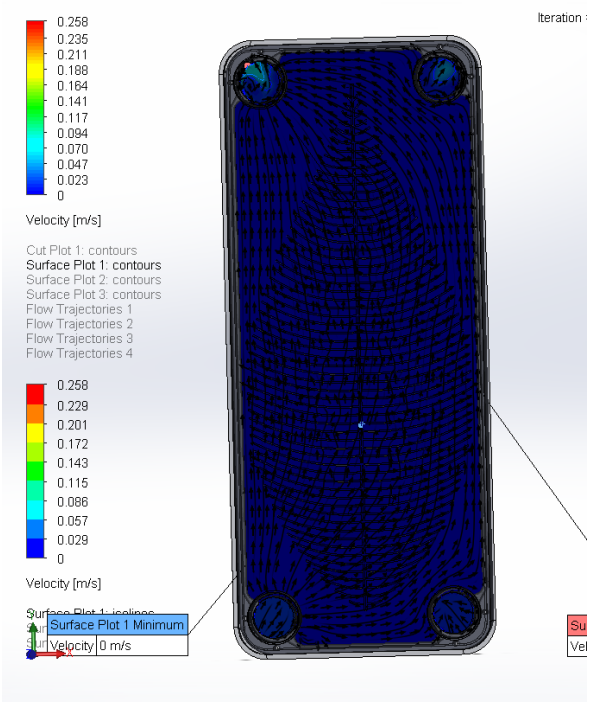


Figure 27. Velocity Contours in Leaf structured PHE

Flow branches from the inlet and spreads across the surface. Velocity is more evenly distributed, avoiding hotspots and dead zones.

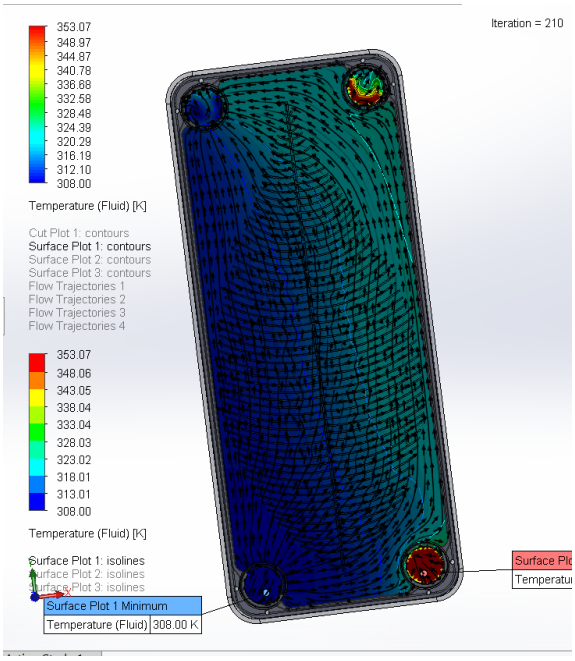


Figure 28. Temperature Contours in Leaf structured PHE

The temperature gradient is well-distributed across the entire plate, showing that flow reaches even the outer regions of the plate, improving overall thermal performance.

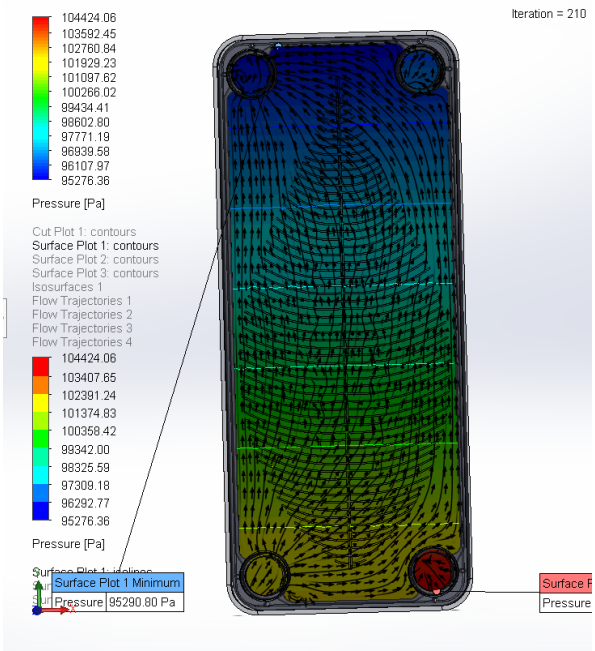


Figure 29. Pressure Contours in Leaf structured PHE

Despite the complex branching, the pressure drop is managed efficiently. Resistance is present but more balanced due to smoother transitions in flow paths.

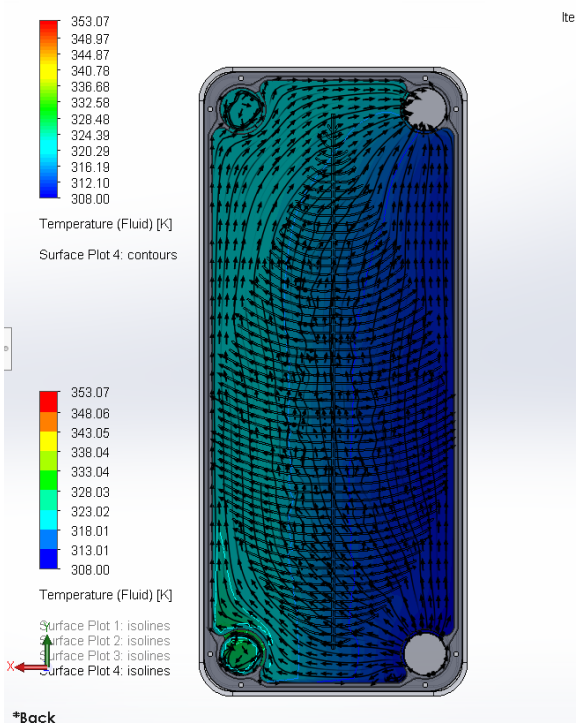


Figure 30. Surface Temperature Plot

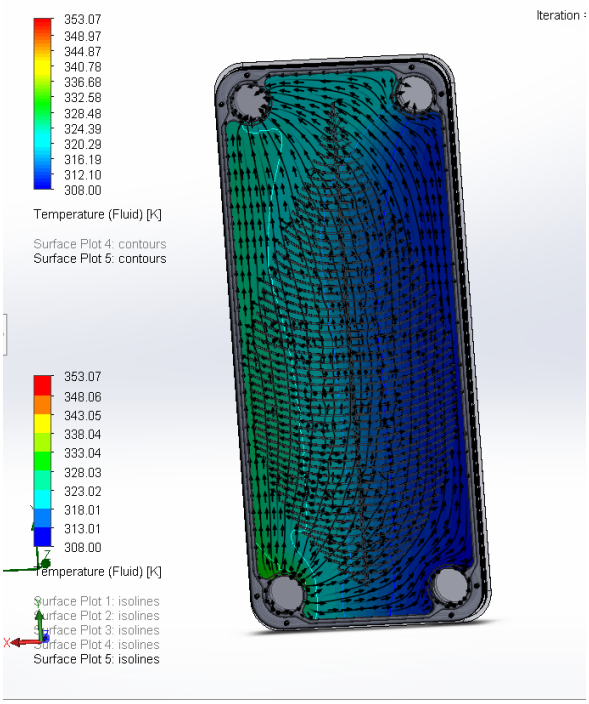


Figure 31. Surface Temperature Plot

Highly uniform surface temperatures indicate efficient usage of the plate's area. This leads to higher thermal effectiveness.

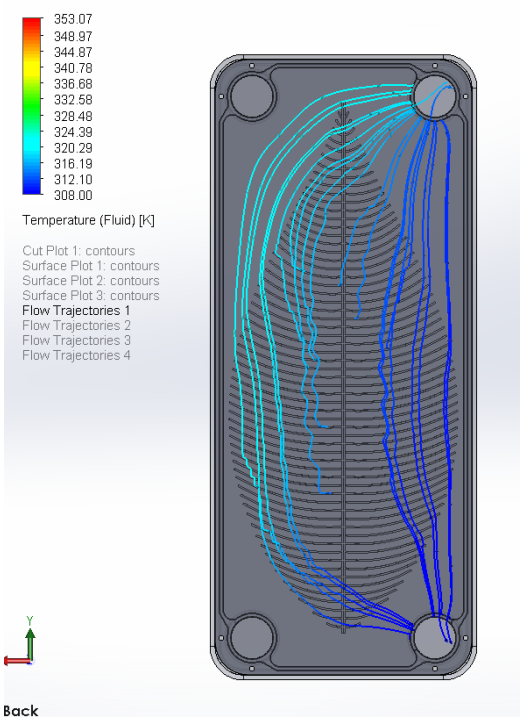


Figure 32. Flow Trajectories

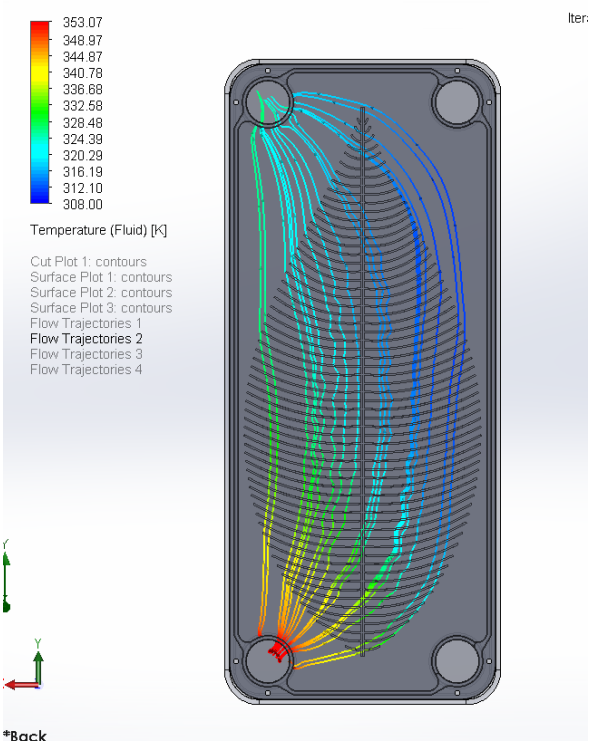


Figure 33. Flow Trajectories

Streamlines divide into multiple branches and smoothly rejoin toward the outlet, ensuring all areas of the heat exchanger participate in the process.

This design provides high heat transfer performance with a surprisingly low pressure drop, making it the most efficient among the tested models.

Table 1. Comparative results of all four types of PHE's

PHE Type	***Q_{oil} [W]***	***Q_{water} [W]***	***Outlet Temp (Oil) [K]***	***Pressure Drop (Oil) [Pa]***	***Change in Pressure Drop [%]***

Plain Plate	571.11	508.23	339.00	9,065.2	0
Grooved (Ribbed) Plate	638.00	541.66	338.01	7,254.3	−19.98
Vein-Structured Plate	737.33	504.88	335.70	22,968.8	+153.45
Leaf-Structured Plate	775.72	656.77	337.59	7,781.8	−14.16

Table 2. Comparison between Heat Transfer Coefficient of all four types of PHE's

	HTC_oil **(W/m²K)**	*HTC_water* **(W/m²K)**	*Overall_htc* **(W/m²K)**	*Percentage_increase(%)*
Plain Type	82.4552	73.37677	77.91599	0
Grooves Type	104.4985	88.71893	96.60874	23.9909019
Vein Type	112.9763	77.35947	95.16787	22.14163871
Leaf Type	117.9192	99.83731	108.8783	39.73802035

The plain (flat) plate heat exchanger served as the control model for evaluating the effect of biomimetic modifications. It delivered an outlet oil temperature of 339 K, indicating the least amount of heat extracted among all configurations. This outcome aligns with its lowest heat transfer coefficient of 77.92 W/m²·K, confirming minimal enhancement of convective heat transfer due to the absence of surface complexity or turbulence-inducing features. However, the flat plate exhibited a pressure drop of only 9,065.2 Pa, highlighting its low flow resistance and energy efficiency from a hydraulic standpoint. This configuration is ideal for applications where pumping power is constrained, and thermal performance is of secondary importance.

The grooved (ribbed) plate, inspired by natural surface corrugations, introduced moderate surface perturbations that improved thermal performance without significantly compromising flow dynamics. It achieved an outlet oil temperature of 338.01 K, and the heat transfer coefficient increased to 96.61 W/m²·K, representing a 23.99% improvement over the flat plate. Interestingly, despite the added surface features, the pressure drop decreased to 7,252.9 Pa—a 20% reduction compared to the plain plate. This suggests that the grooves promoted favorable redirection of the flow, reducing stagnation zones and enhancing heat convection while maintaining an efficient pressure profile. Thus, the grooved plate demonstrates a well-balanced improvement in thermal effectiveness and energy economy, making it suitable for systems requiring modest performance boosts at minimal operational cost.

The vein-structured plate, modeled after natural leaf vein patterns, showed the strongest thermal extraction performance, with the outlet temperature dropping significantly to 335.70 K. This improvement correlates with a heat transfer

coefficient of 95.17 W/m²·K, a 22.14% increase compared to the baseline. The branching vein patterns encouraged localized flow impingement and turbulence, effectively enhancing thermal mixing. However, this geometric complexity came with a notable hydraulic penalty—a pressure drop of 22,968.8 Pa, which is 153.45% higher than the plain plate. This large increase highlights the intense flow resistance created by intricate internal redirection. Therefore, the vein-structured plate is well-suited for high-performance thermal systems, where maximum heat recovery is prioritized over pumping power efficiency.

Finally, the leaf-structured plate offered the most optimal combination of performance metrics. It delivered an outlet oil temperature of 337.59 K, indicating considerable heat extraction. Most notably, it recorded the highest heat transfer coefficient of 108.88 W/m²·K, marking a 39.74% improvement over the flat plate. This substantial gain can be attributed to the bio-inspired branching channels, which guided fluid uniformly across the surface, reducing cold spots and encouraging steady thermal interaction. At the same time, it maintained a low pressure drop of 7,781.8 Pa, which is 14.16% less than the plain plate. The smooth curvature and distributed flow paths in the leaf design contributed to both reduced flow resistance and improved thermal contact, proving that biomimicry can deliver synergistic gains in both heat transfer and energy efficiency. Hence, the leaf plate stands out as the most well-rounded configuration, particularly for applications demanding consistent thermal performance without significantly increasing operating costs.

V. Conclusions

This comparative analysis of four plate heat exchanger (PHE) configurations—plain, grooved, vein-structured, and leaf-structured—highlights the significant impact of biomimetic surface enhancements on thermal and hydraulic performance. The study confirms that the geometric design of flow channels not only dictates heat transfer efficiency but also determines the associated pressure drop and energy consumption. The following key conclusions can be drawn:

1. Plain Plate Configuration: Serving as the baseline, the plain plate offered the lowest heat transfer coefficient (77.92 W/m²·K) and the highest outlet temperature (339 K), indicating minimal thermal performance. However, it featured a relatively low pressure drop of 9,065.2 Pa, making it suitable for applications with tight energy constraints where thermal enhancement is not a priority.

2. Grooved (Ribbed) Plate: This configuration provided a 23.99% improvement in heat transfer coefficient while simultaneously reducing the pressure drop by 20% compared to the flat plate. With a heat transfer coefficient of 96.61 W/m²·K and a moderate outlet temperature of 338.01 K, it offers a balanced performance enhancement ideal for systems prioritizing energy savings with modest thermal improvement.

3. Vein-Structured Plate: With a heat transfer coefficient of 95.17 W/m²·K (22.14% increase) and the lowest outlet temperature of 335.70 K, the vein configuration clearly excelled in thermal performance. However, this came at the cost of a significantly elevated pressure drop (22,968.8 Pa; 153.45% higher), making it viable only for high-demand thermal systems where heat recovery outweighs pumping costs.

4. Leaf-Structured Plate: Demonstrating the highest heat transfer coefficient of 108.88 W/m²·K (39.74% increase) and a notably low pressure drop of 7,781.8 Pa (14.16% reduction), the leaf design presents the most favorable trade-off. With an outlet temperature of 337.59 K, it offers superior heat extraction while preserving hydraulic efficiency. Its

natural flow branching mechanism enhances convective interaction without incurring substantial energy losses, making it the most promising configuration for real-world industrial, sustainable, and energy-optimized applications.

In summary, the study confirms that biomimicry-based enhancements—especially the leaf-inspired design—can significantly boost thermal performance while maintaining or even reducing pressure losses, offering a transformative approach for next-generation heat exchanger development.

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